step 6). An additional amino acid is added to the growing polypeptide, each time mRNA advances through the ribosome. Once a polypeptide chain is assembled, it emerges out from the base of the large subunit (**Fig. 5.13 c**).

Termination is the final phase of translation. Termination of protein synthesis occurs when one of the three stop codons appears in the 'A' site of the ribosome. The terminal codon signals the action of **GTP – dependent release factor,** which cleaves the polypeptide chain from the terminal tRNA releasing it from the translational complex (step 1). The tRNA is then released from the ribosome, which then dissociates into its subunits (step 2) (**Fig. 5.13 d**).

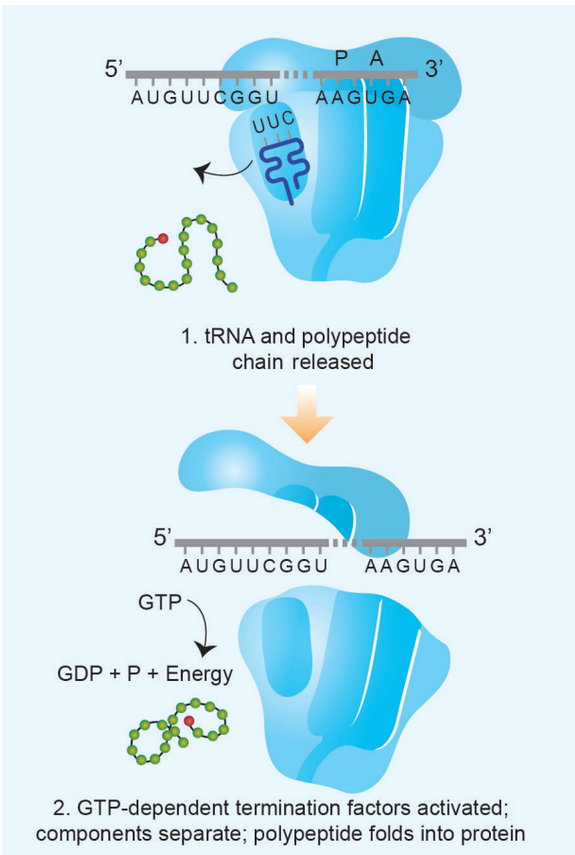


Fig. 5.13 d- Termination of the process of translation

5.13 Regulation of gene expression

We have previously established how DNA is organized into genes, how genes store genetic information, and how this information is expressed. We now consider the most fundamental issues in molecular genetics. How is genetic expression regulated? Evidence in support of the idea that genes can be turned on and off is very convincing. Regulation of gene expression has been extensively studied in prokaryotes, especially in *E. coli*. Gene expression can be controlled or regulated at transcriptional or post transcriptional or translational level. Here, we are going to discuss regulation of gene expression at transcriptional level. Usually, small extracellular or intracellular metabolites trigger initiation or inhibition of gene expression. The clusters of gene with related functions are called **operons**. They usually transcribe single mRNA molecules. In *E.coli*, nearly 260 genes are grouped into 75 different operons.

Many antibiotics do not allow pathogenic bacteria to flourish in animal host because they inhibit one or the other stage of bacterial protein synthesis. The antibiotic tetracycline inhibits binding between aminoacyl tRNA and mRNA. Neomycin inhibits the interaction between tRNA and mRNA. Erythromycin inhibits the translocation of mRNA along the ribosome. Streptomycin inhibits the initiation of translation and causes misreading. Chloramphenicol inhibits peptidyl transferase and formation of peptide bonds.

Structure of the operon: Each operon is a unit of gene expression and regulation and consists of one or more **structural genes** and an adjacent operator gene that controls transcriptional activity of the structural gene.

i) The structural gene codes for proteins, rRNA and tRNA required by the cell.

ii) Promoters are the signal sequences in DNA that initiate RNA synthesis. RNA polymerase binds to the promoter prior to the initiation of transcription.

XII Std Zoology Chapter 5 EM.indd 81 23/12/2022 16:07:16